

IMEN266 Operations Research II

Concept & Proof Companion — Chapter 8 (Queueing theory)

[FULL version — all blanks filled]

Fall 2026

How to use this document

This companion *replaces your class notes*: every definition, model and theorem of the Ch.8 slides (and the derivations developed in class — the stationary distribution, L and W of every queue) is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these eight proofs and their checkpoints are exactly the kind of reasoning the exams reward. Items marked **(final-style)** mirror questions from past finals: the CTMC of a queue with balking and reneging (Proof 5(d), 2025), $M/M/2/K$ from $M/M/2$ by time reversibility (Proof 6(c), 2025), one $M/M/2$ versus two $M/M/1$ queues (Proof 2(c), 2025), the optimal split of a service capacity (Proof 8(e), 2024), and the service rate that keeps the $M/M/1$ sojourn below a target (Proof 4(b), 2024). Slide references are to the 2026 deck (p. n/66).

1 Concept digest

1.1 Kendall notation and the four numbers (slides pp. 1–6/66)

A queueing system is specified by five items — Kendall's $A/B/s/K/D$: the **arrival process** (M Poisson, D deterministic, G/GI general, E_k Erlang), the **service-time law** (same letters), the **number of servers** s , the **capacity** K (waiting space + servers; omitted means ∞) and the **discipline** (FCFS unless said otherwise). $X(t)$ is the number in the system (waiting + in service), $X_q(t)$ the number waiting; W_n , $W_{q,n}$ the sojourn and the waiting time of the n -th customer. The four numbers are two time averages and two customer averages,

$$L = \lim_{t \rightarrow \infty} E[X(t)], \quad L_q = \lim_{t \rightarrow \infty} E[X_q(t)], \quad W = \lim_{n \rightarrow \infty} E[W_n], \quad W_q = \lim_{n \rightarrow \infty} E[W_{q,n}],$$

tied together by **Little's law** $L = \lambda W$, $L_q = \lambda W_q$ (Proof 3) and by the two identities $W = W_q + 1/\mu$, $L = L_q + \lambda/\mu$ (λ/μ = mean number in service = mean number of busy servers). The plan for every $M/M/\cdot$ model: $p_n \rightarrow L = \sum_n n p_n \rightarrow$ the other three. (*verified in notebook: CP11 warm-up: Little's law on a simulated path*)

1.2 Infinite capacity: $M/M/1$, $M/M/2$, $M/M/s$, $M/M/\infty$ (slides pp. 7–22/66)

All four are birth–death CTMCs with birth rate λ and death rate $\mu_n = \min(n, s)\mu$; the **cut equations** $\lambda p_n = \mu_{n+1} p_{n+1}$ give p_n up to p_0 , and normalisation gives p_0 — when the series converges, which is the **stability condition** $\rho = \lambda/(s\mu) < 1$ (Proof 1).

- $M/M/1$: $p_n = (1 - \rho)\rho^n$, $L = \rho/(1 - \rho) = \lambda/(\mu - \lambda)$, $W = 1/(\mu - \lambda)$, $W_q = \lambda/(\mu(\mu - \lambda))$, $L_q = \rho^2/(1 - \rho)$; the sojourn time is $\exp(\mu - \lambda)$ (Proof 4). $L = 1, 9, 99$ at $\rho = 0.5, 0.9, 0.99$: the blow-up as $\rho \uparrow 1$.

- $M/M/2$: $p_0 = \frac{2\mu-\lambda}{2\mu+\lambda}$, $p_n = \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{n-1} p_0$, $L = \frac{4\lambda\mu}{4\mu^2-\lambda^2}$ (Proof 2).
- $M/M/s$: $p_n = \frac{(\lambda/\mu)^n}{n!} p_0$ for $n \leq s$ and $p_s \left(\frac{\lambda}{s\mu}\right)^{n-s}$ beyond; $L_q = \frac{p_s \rho}{(1-\rho)^2}$ with $\rho = \lambda/(s\mu)$; $P(\text{wait}) = p_s/(1-\rho)$ (**Erlang C**).
- $M/M/\infty$: p_n is Poisson with mean λ/μ ; $L = \lambda/\mu$, $W = 1/\mu$, nobody waits; no stability condition; the law survives for any service distribution ($M/G/\infty$).

Two servers of rate μ or one of rate 2μ ? The fast server has the smaller L and W ; the pair has the smaller L_q and W_q (Proof 2(c)). (*verified in notebook: CP11 §1–§4*)

1.3 Finite capacity: $M/M/1/K$, $M/M/s/K$, $M/M/K/K$ (slides pp. 23–30/66)

The chain stops at K : the same cut equations, a *finite* sum, no stability condition. The recipe for every finite-capacity queue (Proof 5):

$$p_K \text{ (loss probability, by PASTA)} \rightarrow \lambda_{\text{eff}} = \lambda(1 - p_K) \rightarrow W = \frac{L}{\lambda(1 - p_K)},$$

then $W_q = W - 1/\mu$ and $L_q = \lambda(1 - p_K)W_q$. $M/M/1/K$: $p_n = \frac{(1-\lambda/\mu)(\lambda/\mu)^n}{1-(\lambda/\mu)^{K+1}}$ (uniform if $\lambda = \mu$). $M/M/K/K$ (Erlang's loss system): $p_n = \frac{a^n/n!}{\sum_{j \leq K} a^j/j!}$, $a = \lambda/\mu$, and p_K is the **Erlang loss formula** (**Erlang B**) — valid for $M/G/K/K$ (insensitivity: only the mean call length enters). Example $M/M/5/5$ with $\lambda = 5/\text{h}$, $1/\mu = 12 \text{ min}$: $p_5 = 1/326$, $L = 325/326$, $W = 12 \text{ min}$. (*verified in notebook: CP12 §1–§3*)

1.4 Bulk arrivals, $M/G/1$ and $G/G/1$ (slides pp. 31–37/66)

$M^{[X]}/M/1$ (batches $\text{PP}(\lambda)$, batch size X , individual $\exp(\mu)$ service): $\rho = \lambda E[X]/\mu$ and $L = \frac{\rho + (\lambda/\mu)E[X^2]}{2(1-\rho)}$; Little's law uses the *customer* rate $\lambda E[X]$ (Proof 7(c)). $M/G/1$ (service mean $1/\mu$, variance σ^2): the **Pollaczek–Khinchine formula**

$$L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}, \quad W_q = \frac{\lambda E[S^2]}{2(1-\rho)}, \quad E[S^2] = \sigma^2 + \frac{1}{\mu^2}$$

— only the mean and the variance of the service time matter; $M/D/1$ has half the queue of $M/M/1$ (Proof 7). $G/G/1$ has no closed form; Appendix A collects the approximations of the PLMS handout. (*verified in notebook: CP12 §5, §6, §8*)

1.5 Time reversibility, Burke's theorem and truncation (slides pp. 38–46/66)

For a stationary CTMC the reversed process is a CTMC with rates $q_{ij}^* = p_j q_{ji}/p_i$; the chain is **time-reversible** if $q_{ij}^* = q_{ij}$, i.e. $p_i q_{ij} = p_j q_{ji}$ (detailed balance). Birth–death chains are reversible (the cut equations *are* the detailed balance equations), so all the $M/M/\cdot$ queues above are. **Burke's theorem**: the departure process of a stationary $M/M/s$ is $\text{PP}(\lambda)$, and the departures before t are independent of $X(t)$. **Truncation theorem**: restricting a reversible chain to $\mathcal{S}' \subset \mathcal{S}$ only renormalises its weights, $p_j^B = p_j^A / \sum_{\mathcal{S}'} p_i^A$ — $M/M/2/K$ from $M/M/2$ (2025 final), $M/M/K/K$ from $M/M/\infty$, two call types sharing K lines from two independent $M/M/\infty$ queues (Proof 6). (*verified in notebook: CP12 §4, CP13 §3*)

1.6 Networks of queues (slides pp. 47–66/66)

An **open Jackson network**: N stations, external $\text{PP}(\theta_i)$ arrivals, s_i servers with $\exp(\mu_i)$ service, infinite capacity, Markovian routing (p_{ij} , exit probability $r_i = 1 - \sum_j p_{ij}$). The state is the vector $X(t) = (X_1(t), \dots, X_N(t))$. Analysis in two steps (Proof 8):

1. **Traffic equations** $\lambda_i = \theta_i + \sum_j \lambda_j p_{ji}$, i.e. $\lambda = \theta(I - P)^{-1}$: the *effective* arrival rate counts every visit; stability needs $\lambda_i < s_i \mu_i$ at every finite-server station.
2. **Product form**: $p(x_1, \dots, x_N) = \prod_i p_i(x_i)$ with p_i the $M/M/s_i$ law with arrival rate λ_i — the station counts are independent at any fixed time. $L = \sum_i L_i$, $W = L / \sum_i \theta_i$ per customer, $W_i = L_i / \lambda_i$ per visit, $W = \sum_i v_i W_i$ with the visit ratios $v_i = \lambda_i / \sum_k \theta_k$.

Examples: the three-station network with feedback ($\lambda = (2.857, 2.857, 0.857)$, $L = 23.25$, $W = 11.6$); Bay-Gull Bagels ($L = 38.4$, $W = 24$ min); the tandem queue ($\lambda_i = \lambda$, $L = \sum \lambda / (\mu_i - \lambda)$); the feedback queue (two Little's laws: $W_{\text{customer}} = W_{\text{node}} / (1 - q)$); Braess' paradox (a free fast link raises everybody's sojourn from $1/(\mu - \lambda) + 2$ to 3 when $\lambda + 1 < \mu \leq 2\lambda + 1$). (verified in notebook: CP13 §1–§6)

1.7 Model zoo

model	p_n (up to p_0)	stable iff	L	remarks
$M/M/1$	ρ^n , $\rho = \lambda/\mu$	$\rho < 1$	$\rho/(1 - \rho)$	sojourn $\exp(\mu - \lambda)$; $W = 1/(\mu - \lambda)$
$M/M/2$	$\frac{\lambda}{\mu} (\frac{\lambda}{2\mu})^{n-1}$	$\lambda < 2\mu$	$\frac{4\lambda\mu}{4\mu^2 - \lambda^2}$	$p_0 = \frac{2\mu - \lambda}{2\mu + \lambda}$
$M/M/s$	$\frac{a^n}{n!}$ ($n \leq s$), $\frac{a^s}{s!} \rho^{n-s}$	$\rho = \frac{a}{s} < 1$	$L_q + a$, $L_q = \frac{ps\rho}{(1-\rho)^2}$	$a = \lambda/\mu$; Erlang C = $p_s/(1 - \rho)$
$M/M/\infty$	$a^n/n!$	always	a	Poisson(a); $W = 1/\mu$; $M/G/\infty$ too
$M/M/1/K$	ρ^n , $n \leq K$	always	$\sum_{n \leq K} np_n$	$W = L/(\lambda(1 - p_K))$
$M/M/s/K$	$M/M/s$ weights, $n \leq K$	always	$\sum np_n$	truncation of $M/M/s$
$M/M/K/K$	$a^n/n!$, $n \leq K$	always	$a(1 - p_K)$	Erlang B = p_K ; $M/G/K/K$ too
$M^{[X]}/M/1$	— (gen. function)	$\lambda E[X] < \mu$	$\frac{\rho + (\lambda/\mu)E[X^2]}{2(1-\rho)}$	Little with $\lambda E[X]$
$M/G/1$	— (embedded DTMC)	$\lambda E[S] < 1$	$L_q + \rho$, $L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$	P-K; $M/D/1$: half of $M/M/1$'s L_q
Jackson	$\prod_i p_i(x_i)$	$\lambda_i < s_i \mu_i$	$\sum_i L_i$	$\lambda = \theta(I - P)^{-1}$; $W = L / \sum \theta_i$

2 Core proofs

Notation: $\text{PP}(\lambda)$ arrivals, $\exp(\mu)$ service unless said otherwise; $p_n = \lim_t P(X(t) = n)$; ρ is the traffic intensity of the model at hand. Blanks are the steps *you* must supply in the gap-fill version.

Proof 1 — Birth–death queues: the cut equations, $M/M/1$ and $M/M/\infty$

Theorem. (a) A birth–death CTMC with rates λ_n (up) and μ_n (down) has stationary distribution $p_n = p_0 \prod_{k=0}^{n-1} \lambda_k / \mu_{k+1}$ provided $\sum_n \prod_{k < n} \lambda_k / \mu_{k+1} < \infty$, and none otherwise. (b) $M/M/1$: if $\rho = \lambda/\mu < 1$ then $p_n = (1 - \rho)\rho^n$, $L = \rho/(1 - \rho)$, $W = 1/(\mu - \lambda)$, $W_q = \lambda/(\mu(\mu - \lambda))$, $L_q = \rho^2/(1 - \rho)$; if $\rho \geq 1$ there is no stationary distribution. (c) $M/M/\infty$: $p_n = e^{-a} a^n / n!$ with $a = \lambda/\mu$, for every λ, μ ; $L = a$, $W = 1/\mu$.

Proof. (a) Global balance at state 0 reads $\lambda_0 p_0 = \mu_1 p_1$. At state $n \geq 1$: $(\lambda_n + \mu_n)p_n = \lambda_{n-1}p_{n-1} + \mu_{n+1}p_{n+1}$. Assume inductively $\lambda_{n-1}p_{n-1} = \mu_n p_n$ (true for $n = 1$); subtracting it from the state- n equation leaves $\lambda_n p_n = \mu_{n+1} p_{n+1}$ — the **cut equation**: in steady state the flow across the cut between n and $n + 1$ balances in the two directions. Iterating, $p_n = \frac{\lambda_{n-1}}{\mu_n} p_{n-1} = \cdots = p_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}$, and p_0 is fixed by $\sum_n p_n = 1$, which is possible iff the series of products converges. (The chain is irreducible, so the stationary distribution, when it exists, is unique and is the limit law.)

(b) $\lambda_n = \lambda$, $\mu_n = \mu$: $p_n = \rho^n p_0$ and $\sum_n \rho^n < \infty$ iff $\rho < 1$; then $p_0 = 1 - \rho$. For L use the derivative trick: for $0 < x < 1$, $\sum_{n \geq 1} n x^{n-1} = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}$, so

$$L = \sum_n n(1 - \rho)\rho^n = (1 - \rho)\rho \sum_n n\rho^{n-1} = \frac{(1 - \rho)\rho}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}.$$

Little (Proof 3) gives $W = L/\lambda = 1/(\mu - \lambda)$; then $W_q = W - 1/\mu = \lambda/(\mu(\mu - \lambda))$ and $L_q = \lambda W_q = \rho^2/(1 - \rho)$; check: $L - L_q = \rho$ = the fraction of time the server is busy = $1 - p_0$. If $\rho \geq 1$ the series diverges: the chain is null recurrent ($\rho = 1$) or transient ($\rho > 1$) and $X(t) \rightarrow \infty$ in probability.

(c) $\lambda_n = \lambda$, $\mu_n = n\mu$: $p_n = p_0 \prod_{k < n} \frac{\lambda}{(k+1)\mu} = p_0 a^n/n!$, and $\sum_n a^n/n! = e^a < \infty$ always, so $p_0 = e^{-a}$: the Poisson law with mean a , hence $L = a$ and $W = L/\lambda = 1/\mu$ — every customer has a server, the sojourn is the service. $\square$

Remark. The cut argument is the whole secret of the chapter: every $M/M/\cdot$ queue below is a birth-death chain, and its p_n is a product of rate ratios read off the diagram. What differs is only the death rate ($\min(n, s)\mu$) and where the diagram stops (K).

(verified in notebook: CP11 §1 ($M/M/1$ help desk, $\rho \rightarrow 1$ run-length table), §4 ($M/M/\infty$, $M/G/\infty$ insensitivity))

Checkpoint — answer before moving on.

- Why does the cut equation hold with equality in steady state but not at every finite t ? (Write $\frac{d}{dt}P(X(t) \leq n)$ in terms of the two flows.)
- The convenience store of HW8: one server at rate 6/h when alone, an assistant raises the rate to 12/h whenever two or more are present, PP(10/h) arrivals. Draw the diagram, write p_n , find L and W . Is it stable? (It is $\lambda < 12$ that matters, not $\lambda < 6$.)
- For $M/M/1$ with $\rho = 0.99$, $L = 99$. How long must a simulation run to estimate L to $\pm 10\%$? (The relaxation time scales like $1/(1 - \rho)^2$ — CP11 §1's table.)

AI study card — paste one of these into your AI assistant:

- Ask me to derive the stationary distribution of a general birth-death chain from the global balance equations by induction (the cut equations), and then to specialise it to $M/M/1$ and $M/M/\infty$. Check each step before showing the next.
- Quiz me on the $M/M/1$ queue: give me ρ and ask for L , W , L_q , W_q and the busy fraction; then ask what happens to each as ρ goes to 1 and why a stationary distribution fails to exist at $\rho = 1$.

Proof 2 — $M/M/s$: p_0 , L_q and Erlang C; $M/M/2$; two servers or one fast server (final-style)

Theorem. Let $a = \lambda/\mu$ and $\rho = \lambda/(s\mu) < 1$. (a) $p_n = \frac{a^n}{n!}p_0$ for $0 \leq n \leq s$, $p_n = p_s \rho^{n-s}$ for $n \geq s$, and $p_0 = \left[\sum_{n=0}^{s-1} \frac{a^n}{n!} + \frac{a^s}{s!} \frac{1}{1-\rho} \right]^{-1}$. (b) $L_q = \frac{p_s \rho}{(1 - \rho)^2}$, $W_q = L_q/\lambda$, $W = W_q + 1/\mu$, $L = L_q + a$,

and $P(\text{an arrival waits}) = \sum_{n \geq s} p_n = \frac{p_s}{1-\rho}$ (Erlang C). (c) $M/M/2$: $p_0 = \frac{1-\rho}{1+\rho}$ with $\rho = \lambda/(2\mu)$, $L = \frac{2\rho}{1-\rho^2} = \frac{4\lambda\mu}{4\mu^2 - \lambda^2}$, $L_q = \frac{2\rho^3}{1-\rho^2}$. Compared with one server of rate 2μ (an $M/M/1$ with the same ρ , $L' = \frac{\rho}{1-\rho}$, $L'_q = \frac{\rho^2}{1-\rho}$): $L/L' = \frac{2}{1+\rho} > 1$ but $L_q/L'_q = \frac{2\rho}{1+\rho} < 1$.

Proof. (a) The death rate is $\min(n, s)\mu$. Below s the cut equations give $p_n = \frac{\lambda}{n\mu} p_{n-1}$, so $p_n = \frac{a^n}{n!} p_0$; from s on the ratio is constant, $p_{n+1} = \frac{\lambda}{s\mu} p_n = \rho p_n$, so $p_{s+i} = \rho^i p_s$. Normalising, $1 = p_0 \sum_{n < s} \frac{a^n}{n!} + p_s(1 + \rho + \rho^2 + \dots)$, and the geometric tail converges iff $\rho < 1$, giving p_0 as stated (with $p_s = \frac{a^s}{s!} p_0$). (b) Only the states above s hold a queue, of length $n - s$:

$$L_q = \sum_{i \geq 1} i p_{s+i} = p_s \sum_{i \geq 1} i \rho^i = p_s \rho \sum_{i \geq 1} i \rho^{i-1} = \frac{p_s \rho}{(1-\rho)^2}$$

by the derivative trick. $W_q = L_q/\lambda$ and $W = W_q + 1/\mu$ are Little's law and the service identity; $L = \lambda W = L_q + a$. An arrival (PASTA, Proof 4) waits iff it finds all s servers busy: $P(\text{wait}) = \sum_{n \geq s} p_s \rho^{n-s} = p_s/(1-\rho)$.

(c) With $s = 2$, $a = 2\rho$: $p_0 = [1 + 2\rho + \frac{(2\rho)^2}{2} \frac{1}{1-\rho}]^{-1} = [1 + 2\rho + \frac{2\rho^2}{1-\rho}]^{-1} = \frac{1-\rho}{(1+2\rho)(1-\rho)+2\rho^2} = \frac{1-\rho}{1+\rho}$ (this is the deck's $\frac{2\mu-\lambda}{2\mu+\lambda}$). Then $p_2 = 2\rho^2 p_0$, $L_q = \frac{2\rho^3 p_0}{(1-\rho)^2} = \frac{2\rho^3}{(1-\rho)(1+\rho)} = \frac{2\rho^3}{1-\rho^2}$ and $L = L_q + 2\rho = \frac{2\rho^3+2\rho(1-\rho^2)}{1-\rho^2} = \frac{2\rho}{1-\rho^2}$; substituting $\rho = \lambda/(2\mu)$ gives $\frac{4\lambda\mu}{4\mu^2 - \lambda^2}$. For the single fast server $L' = \rho/(1-\rho)$, so $L/L' = \frac{2\rho}{1-\rho^2} \cdot \frac{1-\rho}{\rho} = \frac{2}{1+\rho} \in (1, 2)$: the fast server always holds fewer customers (every service takes half as long). But $L_q/L'_q = \frac{2\rho^3}{1-\rho^2} \cdot \frac{1-\rho}{\rho^2} = \frac{2\rho}{1+\rho} < 1$: fewer customers *wait* with two servers, because with one customer present the pair still has a free server while the fast server is busy with him. $\square$

Remark (final-style, 2025 Q4: pool or split?). Two streams $PP(\lambda)$ each, two servers of rate μ , $\rho = \lambda/\mu < 1$. *Split* (each stream its own $M/M/1$): $L_{\text{split}} = \frac{2\rho}{1-\rho}$. *Pooled* (one $M/M/2$ fed by $PP(2\lambda)$, so the same ρ per server): $L_{\text{pool}} = \frac{2\rho}{1-\rho^2}$. Hence $L_{\text{pool}}/L_{\text{split}} = \frac{1}{1+\rho} < 1$ and, by Little with the same total rate 2λ , $W_{\text{pool}}/W_{\text{split}} = \frac{1}{1+\rho}$ too. Intuition: in the split system a server can be idle while a customer waits in the other line; the pooled queue never wastes capacity that way. ($\rho = 2/3$: the ratio is 0.6.)

(verified in notebook: CP11 §2 (two servers vs one fast), §3 ($M/M/5$ bank, Erlang C); CP13 §5 (pooling))

Checkpoint — answer before moving on.

- **(final-style)** Compare one $M/M/2$ with two $M/M/1$ queues fed by a $PP(2\lambda)$ split in halves, in L and W , and explain the intuition. Then redo the comparison for L_q and W_q — is the ranking the same?
- HW8's bank: $\lambda = 40/\text{h}$, $\mu = 10/\text{h}$, $s = 5$. Compute p_0 , $P(\text{wait})$, L_q , W_q . How many tellers are needed for $P(\text{wait}) \leq 0.2$?
- Show that $P(\text{wait}) = p_s/(1-\rho)$ equals $p_s + p_{s+1} + \dots$ and explain in words why it is larger than p_s .

AI study card — paste one of these into your AI assistant:

- Derive p_0 and L_q of the $M/M/s$ queue from the two regimes of the birth-death diagram, asking me for each ratio before you reveal it. Then ask me to specialise to $s = 2$ and to compare with an $M/M/1$ of rate 2μ in L and in L_q .
- Give me a staffing problem: arrivals 40/h, service 10/h per teller, target probability of waiting 0.2. Make me compute Erlang C for $s = 5, 6, 7$ by hand and check my numbers.

Proof 3 — Little’s law $L = \lambda W$ as a renewal reward identity, and the two identities

Theorem. For a queue in which the arrival rate $\lambda = \lim_t A(t)/t$ exists ($A(t)$ = arrivals in $[0, t]$) and the limits L and W exist, $L = \lambda W$. The same holds for the queue alone ($L_q = \lambda W_q$) and for the servers alone (mean number in service = $\lambda E[S] = \lambda/\mu$), hence $L = L_q + \lambda/\mu$ and $W = W_q + 1/\mu$. In a loss system the λ to use is the rate of customers who *enter*, $\lambda_{\text{eff}} = \lambda(1 - p_K)$.

Proof (sample path). Let customer n arrive at a_n and leave at d_n , with sojourn $W_n = d_n - a_n$. At time t the number present is $X(t) = \sum_n \mathbf{1}\{a_n \leq t < d_n\}$. Integrate over $[0, T]$ and exchange sum and integral:

$$\int_0^T X(t) dt = \sum_n |[a_n, d_n) \cap [0, T]| = \sum_{n: d_n \leq T} W_n + \sum_{n: a_n \leq T < d_n} (T - a_n) \leq \sum_{n \leq A(T)} W_n.$$

Divide by T and write the right side as $\frac{A(T)}{T} \cdot \frac{1}{A(T)} \sum_{n \leq A(T)} W_n \rightarrow \lambda W$. For the reverse inequality, $\int_0^T X dt \geq \sum_{n: d_n \leq T} W_n$, and the customers who have left by T are at least $A(T) - X(T)$ in number; when $X(T)/T \rightarrow 0$ (true whenever $L < \infty$ along a regenerative sequence of times, e.g. the empty epochs), both bounds give $L = \lambda W$.

Proof (renewal reward, Ch. 7). Suppose the queue empties again and again (the empty epochs are regeneration points). In one cycle let N be the number of customers served and $R = \int_{\text{cycle}} X(t) dt = \sum_{n \in \text{cycle}} W_n$ the area under the path. By the renewal reward theorem, $L = \frac{E[R]}{E[\text{cycle}]}$, $\lambda = \frac{E[N]}{E[\text{cycle}]}$, and $W = \frac{E[\sum_{n \in \text{cycle}} W_n]}{E[N]} = \frac{E[R]}{E[N]}$. Therefore $\lambda W = \frac{E[N]}{E[\text{cycle}]} \cdot \frac{E[R]}{E[N]} = \frac{E[R]}{E[\text{cycle}]} = L$.

The two identities. Apply the law to the box “servers”: a customer is in the box for its service time, so the mean number in service equals $\lambda E[S] = \lambda/\mu$ (for one server this is the busy fraction $\rho = 1 - p_0$); $X = X_q + (\text{number in service})$ gives $L = L_q + \lambda/\mu$, and $W_n = W_{q,n} + S_n$ gives $W = W_q + 1/\mu$. In a finite-capacity system only entering customers have a sojourn; they arrive at rate $\lambda(1 - p_K)$, which is the λ of every Little identity there. $\square$

Remark. Nothing about Poisson arrivals or exponential service was used: Little’s law is a bookkeeping identity between a time average and a customer average, valid for any discipline. What the service law and the arrival law decide is the *values* of L and W , never the relation between them.

(verified in notebook: CP11 warm-up (area under the path vs sum of sojourns); CP12 §1 (effective rate: $W = 25$ min, not 20))

Checkpoint — answer before moving on.

- HW8’s barber shop: $p_0 = \frac{1}{16}$, $p_1 = \frac{4}{16}$, $p_2 = \frac{6}{16}$, $p_3 = \frac{4}{16}$, $p_4 = \frac{1}{16}$, two barbers, 4 customers per hour *enter and stay*. Compute L , L_q , W , W_q and the mean haircut time. (Which arrival rate goes into Little’s law here, and why is the mean haircut time not $1/\mu$ of a formula?)
- In the car wash of CP12 §1 ($M/M/1/4$, $\lambda = \mu$) the naive L/λ gives 20 min and the correct W is 25 min. Which customers does the naive version pretend to count?
- Little’s law for the servers of an $M/M/s$ queue gives $L - L_q = \lambda/\mu$. Verify it from the formulas of Proof 2 for $s = 2$.

AI study card — paste one of these into your AI assistant:

- Ask me to prove Little’s law from a sample path: the area under the number-in-system curve equals the sum of the sojourn times. Then ask me to redo it with the renewal reward theorem using the empty epochs as regeneration points.
- Give me three queueing systems (one with blocking) and ask which arrival rate goes into $L = \lambda W$ in each. Make me explain the mistake of dividing L by the raw arrival rate in a loss system.

Proof 4 — PASTA, and the $M/M/1$ sojourn time is $\exp(\mu - \lambda)$ (final-style)

Theorem. (a) (PASTA: Poisson Arrivals See Time Averages.) If arrivals form a $PP(\lambda)$ and the future arrivals after t are independent of the history of the system up to t (“lack of anticipation”), then the long-run fraction of arrivals that find the system in a set of states B equals the long-run fraction of time the system spends in B . In particular an arrival finds n customers with probability p_n . (b) In a stationary $M/M/1$ queue the sojourn time of a customer is $\exp(\mu - \lambda)$; its waiting time is 0 with probability $1 - \rho$ and $\exp(\mu - \lambda)$ with probability ρ . (c) (2024 final Q2) If the system must satisfy $\lim_n P(W_n > x) \leq \varepsilon$, the cheapest service rate is $\mu^* = \lambda + \frac{1}{x} \ln \frac{1}{\varepsilon}$.

Proof. (a) Fix B and let $Z(t) = \mathbf{1}\{X(t) \in B\}$. Cut $[0, t]$ into m pieces of length $h = t/m$. The number of arrivals in $(s, s + h]$ that find the system in B is, up to $o(h)$, $Z(s) \cdot \mathbf{1}\{\text{arrival in } (s, s + h]\}$, and by lack of anticipation the arrival indicator is independent of $Z(s)$, with mean $\lambda h + o(h)$. Summing over the pieces,

$$E[\#\{\text{arrivals in } [0, t] \text{ finding } B\}] = \lim_{m \rightarrow \infty} \sum_{k < m} E[Z(kh)] \lambda h = \lambda \int_0^t E[Z(s)] ds = \lambda E\left[\int_0^t Z(s) ds\right].$$

Divide by $E[A(t)] = \lambda t$: the expected fraction of arrivals finding B equals the expected fraction of time in B ; the ergodic theorem turns both expectations into long-run fractions. The independence is essential: with arrivals every 1 time unit and services of length $\frac{1}{2}$, every arrival finds an empty system although the server is busy half the time.

(b) An arrival finds N customers, with $P(N = n) = (1 - \rho)\rho^n$ by PASTA. It must wait for N full services (the residual service of the customer being served is again $\exp(\mu)$ by memorylessness) and then its own: the sojourn is a sum of $N + 1$ iid $\exp(\mu)$ variables with $N + 1$ geometric, $P(N + 1 = k) = (1 - \rho)\rho^{k-1}$. A geometric sum of exponentials is exponential: by conditioning on the first phase and on whether another phase follows, or by transforms,

$$E[e^{-sT}] = \sum_{k \geq 1} (1 - \rho)\rho^{k-1} \left(\frac{\mu}{\mu + s}\right)^k = \frac{(1 - \rho)\mu}{\mu + s - \rho\mu} = \frac{\mu - \lambda}{\mu - \lambda + s},$$

the transform of $\exp(\mu - \lambda)$. The waiting time is the same sum without the last phase: 0 when $N = 0$, otherwise a geometric number (≥ 1) of phases, again $\exp(\mu - \lambda)$.

(c) $P(W_n > x) \rightarrow e^{-(\mu - \lambda)x} \leq \varepsilon$ iff $\mu \geq \lambda + \frac{1}{x} \ln \frac{1}{\varepsilon}$; since a lower rate costs less, $\mu^* = \lambda + \frac{1}{x} \ln(1/\varepsilon)$. $\square$

Remarks. (i) PASTA is why p_K is the loss probability of a finite-capacity queue and why $P(\text{wait}) = \sum_{n \geq s} p_n$ in $M/M/s$: the “time-average” p_n doubles as the “arriving customer’s” p_n . (ii) The service rate in (c) grows only logarithmically in $1/\varepsilon$: halving the tolerance costs a fixed increment $\ln 2/x$.

(verified in notebook: CP11 §1 ($P(\text{sojourn} > 2h) = e^{-2}$; μ^* for $\varepsilon = 5\%$), §5 (PASTA for Poisson, deterministic and Erlang arrivals))

Checkpoint — answer before moving on.

- **(final-style)** Given that the stationary $M/M/1$ sojourn is $\exp(\mu - \lambda)$, derive the optimal μ for $\lim_n P(W_n > x) \leq \varepsilon$. Then explain why the result does not change if “sojourn” is replaced by “waiting time” and $\varepsilon < \rho$.
- Deterministic arrivals every 2 min to a server with $\exp(1/\text{min})$ service: is the fraction of arrivals that wait larger or smaller than the busy fraction? (CP11 §5 shows the direction; argue it without simulation.)
- Where exactly does the proof of (b) use memorylessness, and what breaks for $M/G/1$?

AI study card — paste one of these into your AI assistant:

- Explain PASTA to me with the lack-of-anticipation argument, then give me a counterexample with deterministic arrivals and make me compute the time-average and the arrival-average busy fractions.
- Ask me to show that a geometric sum of iid exponential(μ) variables is exponential, and to use it for the sojourn time of the stationary $M/M/1$ queue. Then pose the 2024 final question about the cheapest μ meeting a tail constraint.

Proof 5 — Finite capacity: $M/M/1/K$, $M/M/s/K$, Erlang’s loss formula, and a queue with balking and reneging (final-style)

Theorem. (a) $M/M/1/K$: $p_n = \frac{(1-\rho)\rho^n}{1-\rho^{K+1}}$, $n = 0, \dots, K$ ($\rho = \lambda/\mu \neq 1$; $p_n = \frac{1}{K+1}$ if $\rho = 1$), for every ρ ; p_K is the fraction of arrivals lost and $W = L/(\lambda(1-p_K))$. (b) $M/M/s/K$: the $M/M/s$ weights of Proof 2 restricted to $n \leq K$ and renormalised. (c) $M/M/K/K$: $p_n = \frac{a^n/n!}{\sum_{j=0}^K a^j/j!}$, $a = \lambda/\mu$; the loss probability $B(K, a) = p_K$ (**Erlang B**) satisfies $B(K, a) = \frac{aB(K-1, a)}{K+aB(K-1, a)}$, $B(0, a) = 1$; $L = a(1-B)$ and $W = 1/\mu$. The formula holds for $M/G/K/K$ with mean service time $1/\mu$ (insensitivity). (d) (2025 final Q2) A single $\exp(\mu)$ server, potential customers $\text{PP}(\lambda)$, an arrival finding i in the system balks with probability $i/(i+1)$, and each waiting customer reneges after an $\exp(\theta)$ patience: the number in system is a birth–death chain with $\lambda_i = \frac{\lambda}{i+1}$ and $\mu_i = \mu + (i-1)\theta$ ($i \geq 1$); it is stable for every λ , with $p_i = p_0 \lambda^i / (i! \prod_{k=0}^{i-1} (\mu + k\theta))$.

Proof. (a) Same cut equations as $M/M/1$, $p_n = \rho^n p_0$, but only for $n \leq K$: $p_0 \sum_{n=0}^K \rho^n = 1$ gives $p_0 = \frac{1-\rho}{1-\rho^{K+1}}$, a finite sum — no condition on ρ . By PASTA an arrival finds the system full with probability p_K and is lost; the entering customers arrive at rate $\lambda(1-p_K)$, which is the rate in Little’s law (Proof 3): $W = L/(\lambda(1-p_K))$, $W_q = W - 1/\mu$, $L_q = \lambda(1-p_K)W_q$. (b) Identical: the diagram of $M/M/s$ cut at K (for $s \leq K$); $L_q = \sum_{n>s} (n-s)p_n$ directly, and $L - L_q = \lambda(1-p_K)/\mu$ is the mean number of busy servers (Little for the servers). (c) $\mu_n = n\mu$ for $n \leq K$: $p_n = \frac{a^n}{n!} p_0$, normalised over $0..K$. For the recursion write $S_K = \sum_{j \leq K} a^j/j!$, so $B(K) = \frac{a^K/K!}{S_K}$ and $S_K = S_{K-1} + \frac{a^K}{K!}$. Then

$$\frac{1}{B(K)} = \frac{S_{K-1}}{a^K/K!} + 1 = \frac{K}{a} \cdot \frac{S_{K-1}}{a^{K-1}/(K-1)!} + 1 = \frac{K}{a B(K-1)} + 1,$$

which rearranges to the claim; it is numerically stable where $a^K/K!$ over- or underflows. $L = \sum_n n p_n = a \sum_{n \geq 1} \frac{a^{n-1}}{(n-1)!} p_0 = a(1-p_K)$ (shift the index), the carried load; nobody waits, so $W = 1/\mu$ — consistent with Little, $W = L/(\lambda(1-p_K)) = a/\lambda = 1/\mu$. Insensitivity is stated without proof (it follows from the reversibility of the $M/G/K/K$ process with the residual service times as

supplementary state, Ross §8.x); CP12 §3 checks it by simulation with exponential, constant and lognormal call lengths.

(d) In state i a potential arrival ($PP(\lambda)$) joins with probability $1 - \frac{i}{i+1} = \frac{1}{i+1}$: Poisson thinning with a state-dependent probability gives the birth rate $\lambda_i = \lambda/(i+1)$. Deaths: the service completes at rate μ (if $i \geq 1$) and each of the $i-1$ *waiting* customers reneges at rate θ , independently — the minimum of independent exponentials — so $\mu_i = \mu + (i-1)\theta$. By Proof 1(a), $p_i = p_0 \prod_{k=0}^{i-1} \frac{\lambda/(k+1)}{\mu+k\theta} = p_0 \frac{\lambda^i}{i! \prod_{k<i} (\mu+k\theta)}$; the ratio $\lambda_i/\mu_{i+1} = \frac{\lambda}{(i+1)(\mu+i\theta)} \rightarrow 0$, so the series converges for every λ (even with $\theta = 0$, where p_i is $Poisson(\lambda/\mu)$: balking alone tames the queue). $\square$

Remarks. (i) The Example $M/M/5/5$ of the deck ($a = 1$): $p_5 = \frac{1/120}{326/120} = \frac{1}{326}$, $L = \frac{325}{326}$, $W = 12$ min; dividing L by λ instead of $\lambda(1 - p_5)$ gives 11.96 min — a sojourn shorter than a service, the sign of a forgotten p_K . (ii) The telephone company of HW9 earns $ca(1 - B(K)) - dK$ per unit time with K lines (c per busy line, d per leased line): B is convex in K , so each extra line earns less than the previous one. (iii) In (d) the long-run fractions of potential customers are: served $= \mu(1 - p_0)/\lambda$ (completions per unit time over potential arrivals), balking $= \sum_i p_i \frac{i}{i+1}$, reneging $= \theta \sum_{i \geq 2} (i-1)p_i/\lambda$; they add up to 1 because the entering rate $\lambda \sum_i p_i/(i+1)$ equals served + reneged in steady state (CP12 §7: $\lambda = 2$, $\mu = 1$, $\theta = 0.5$ give $L = 1.32$, 47% balk, 13% renege, 40% served).

(verified in notebook: CP12 §1 (car wash), §2 ($M/M/5/5$, $M/M/2/4$), §3 (Erlang B , revenue, insensitivity), §7 (balking + reneging: $L = 1.32$, 47% balk, 13% renege))

Checkpoint — answer before moving on.

- **(final-style)** Model the single-server queue with $\exp(\theta)$ patience and balking probability $i/(i+1)$ as a CTMC: state space, rate diagram, all rates, and the balance equation of state i . Then explain why it is stable for every λ .
- HW9's fast food: $PP(3/\text{min})$, two servers with 30-s service, room for 4: compute L and W of the $M/M/2/4$ queue (answer: loss 12.4%, $L = 1.73$, $W = 0.66$ min).
- Prove $L = a(1 - B)$ for $M/M/K/K$ in one line by Little's law applied to the lines.

AI study card — paste one of these into your AI assistant:

- Walk me through the $M/M/1/K$ queue: stationary distribution for rho different from 1 and for rho = 1, the loss probability, and why Little's law must use lambda (1 - p.K). Then ask me to redo the car wash example with $K = 6$.
- Give me a queue with balking probability $i/(i+1)$ and exponential reneging. Ask me for the birth and death rates, the stationary distribution, and the long-run fractions of customers served, balking and reneging. Check my reasoning for the factor $(i - 1)$.

Proof 6 — Time reversibility: the reversed chain, Burke's theorem, and the truncation theorem (final-style)

Theorem. Let X be a stationary ergodic CTMC with rates q_{ij} and stationary law p . (a) The reversed process $X^*(t) = X(T - t)$ is a CTMC with rates $q_{ij}^* = p_j q_{ji}/p_i$; X is **time-reversible** ($q^* = q$) iff $p_i q_{ij} = p_j q_{ji}$ for all i, j . (b) Every birth-death chain is time-reversible. (Burke) In a stationary $M/M/s$ ($s \leq \infty$) the departure process is $PP(\lambda)$, and the departures before t are independent of $X(t)$. (c) (Truncation) If X is reversible on $\mathcal{S}$ and B is the chain on $\mathcal{S}' \subset \mathcal{S}$ with the same rates between states of $\mathcal{S}'$ (arcs leaving $\mathcal{S}'$ deleted, B irreducible), then B is reversible with

$p_j^B = p_j^A / \sum_{i \in \mathcal{S}'} p_i^A$, $j \in \mathcal{S}'$. In particular $M/M/2/K$ has $p_j^{(K)} \propto \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{j-1}$ ($1 \leq j \leq K$), $p_0^{(K)} \propto 1$, and $M/M/K/K$ is the Poisson law of $M/M/\infty$ truncated to $\{0, \dots, K\}$.

Proof. (a) Looking backwards from time t ,

$$P_{ij}^*(t) = P(X(0) = j \mid X(t) = i) = \frac{P(X(t) = i \mid X(0) = j) P(X(0) = j)}{P(X(t) = i)} = \frac{P_{ji}(t) p_j}{p_i},$$

so $p_i P_{ij}^*(t) = p_j P_{ji}(t)$; the reversed process is Markov because the past of a Markov process given its present is independent of its future, which is symmetric in time. Divide by t and let $t \downarrow 0$ ($P_{ji}(t)/t \rightarrow q_{ji}$ for $j \neq i$): $p_i q_{ij}^* = p_j q_{ji}$. Reversible means $q_{ij}^* = q_{ij}$, i.e. $p_i q_{ij} = p_j q_{ji}$ (**detailed balance**).

(b) In a birth–death chain the only arcs are $n \leftrightarrow n+1$, and detailed balance on them is $\lambda_n p_n = \mu_{n+1} p_{n+1}$ — exactly the cut equations of Proof 1, which the stationary law satisfies. Hence the reversed $M/M/s$ is again an $M/M/s$ with the same rates and the same stationary law. Departures of the forward process are arrivals of the reversed one, hence a $PP(\lambda)$. For the independence: in the reversed process the arrivals *after* (reversed) time t are independent of the state at t (Poisson arrivals do not anticipate); read forwards, these are the departures *before* t , and the state is $X(t)$: so **departures before $t \perp X(t)$** .

(c) Candidate: $p_j^B = p_j^A / \sum_{\mathcal{S}'} p_i^A$. For $i, j \in \mathcal{S}'$ the rates are unchanged and A is reversible, so $p_i^A q_{ij} = p_j^A q_{ji}$; dividing by the constant $\sum_{\mathcal{S}'} p_i^A$ gives $p_i^B q_{ij} = p_j^B q_{ji}$ — detailed balance for B . Summing over $i \in \mathcal{S}'$: $\sum_i p_i^B q_{ij} = p_j^B \sum_i q_{ji}$, i.e. the global balance $p^B Q_B = 0$ (for B the row sums run over $\mathcal{S}'$ only, since the deleted arcs are gone from both sides). So p^B is the stationary law of B and B is reversible. $M/M/2/K$: the $M/M/2$ weights (Proof 2, or the deck's $p_j \propto \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{j-1}$) renormalised on $\{0, \dots, K\}$:

$$p_0^{(K)} = \left[1 + \frac{\lambda}{\mu} \sum_{i=1}^K \left(\frac{\lambda}{2\mu}\right)^{i-1}\right]^{-1} = \left[1 + \frac{\lambda}{\mu} \cdot \frac{1 - (\lambda/2\mu)^K}{1 - \lambda/2\mu}\right]^{-1}, \quad p_j^{(K)} = \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{j-1} p_0^{(K)}.$$

Only ratios of weights enter, so $\lambda < 2\mu$ is not needed. For $\lambda = \mu$ and $K = 4$ this gives $p^{(4)} = (0.348, 0.348, 0.174, 0.087, 0.043)$. $M/M/K/K$: truncate the Poisson weights $a^j/j!$ of $M/M/\infty$ to $j \leq K$ — Erlang's distribution in one line. $\square$

Remarks. (i) Reversibility is essential in (c): for the one-way cycle $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ (rates 1), p^A is uniform, but the chain truncated to $\{0, 1\}$ is absorbed in 1 — $p^B = (0, 1)$, not $(\frac{1}{2}, \frac{1}{2})$ (CP12 §4). (ii) Any $\mathcal{S}'$ works: two independent $M/M/\infty$ queues restricted to $x_1 + x_2 \leq K$ (two call types sharing K lines) have $p^B(x_1, x_2) \propto \frac{a_1^{x_1}}{x_1!} \frac{a_2^{x_2}}{x_2!}$ on the triangle, without a single balance equation. (iii) Burke's theorem is the engine of Proof 8: a station fed by a Poisson stream feeds the next one with a Poisson stream, and the independence clause is what makes the counts at a fixed time independent (CP13 §3).

(verified in notebook: CP12 §4 ($M/M/2/4$ by truncation vs the direct solution; the cycle counterexample); CP13 §3 (Burke: inter-departure times, lag correlations))

Checkpoint — answer before moving on.

- (**final-style**) Calculate the steady-state probabilities of $M/M/2/K$ ($K > 2$) using time reversibility and the $M/M/2$ result of the lecture — without solving $pQ = 0$. State the theorem you use and check the hypothesis it needs.
- Show that the $M/M/1$ queue is reversible by checking detailed balance directly, and deduce that an observer who watches only the departures cannot tell whether the film runs forwards or backwards.

- Is the $M/M/1$ queue with *two* customer classes and priority reversible? (Hint: is the number-in-system process even Markov on its own?)

AI study card — paste one of these into your AI assistant:

- Ask me to derive the transition rates of the time-reversed stationary CTMC and to show that birth-death chains are time-reversible. Then make me prove Burke's theorem from reversibility, including the independence of past departures and the present state.
- Pose the truncation theorem for reversible CTMCs and ask me to prove it. Then give me $M/M/2/5$ with $\lambda = 3$, $\mu = 2$ and ask me to compute the stationary distribution from the $M/M/2$ weights only, and to say why the stability condition is irrelevant.

Proof 7 — $M/G/1$: the Pollaczek–Khinchine formula from the residual service time, and bulk arrivals

Theorem. (a) $M/G/1$, FCFS, service S with $E[S] = 1/\mu$, $\text{Var}(S) = \sigma^2$, $\rho = \lambda E[S] < 1$:

$$W_q = \frac{\lambda E[S^2]}{2(1-\rho)}, \quad L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}, \quad W = W_q + \frac{1}{\mu}, \quad L = L_q + \rho.$$

(b) At a fixed ρ , L_q is affine in the squared coefficient of variation $c_s^2 = \sigma^2 \mu^2$: $L_q = \frac{\rho^2(1+c_s^2)}{2(1-\rho)}$; $M/D/1$ has half the queue of $M/M/1$. (c) $M^{[X]}/M/1$: batches $PP(\lambda)$, $P(X = i) = q_i$, individual $\exp(\mu)$ service, $\rho = \lambda E[X]/\mu < 1$: $p_0 = 1 - \rho$ and $L = \frac{\rho + (\lambda/\mu)E[X^2]}{2(1-\rho)}$.

Proof. (a) (mean-value argument). An arriving customer waits for (i) the *residual* service time R of the customer in service, if any, and (ii) the full services of the L_q customers ahead of it (FCFS; PASTA says the arrival sees the time-average queue): $W_q = E[R] + L_q E[S]$. For $E[R]$ use Ch. 7: each completed service of length S contributes the triangle $\int_0^S R dt = S^2/2$ to the area under the residual-time curve, and services are completed at rate λ in the long run, so the time average is $E[R] = \lambda E[S^2]/2$ (equivalently: the server is busy a fraction ρ of the time, and during busy time the residual averages $E[S^2]/(2E[S])$, the inspection-paradox value). With Little, $L_q = \lambda W_q$:

$$W_q = \frac{\lambda E[S^2]}{2} + \rho W_q \implies W_q = \frac{\lambda E[S^2]}{2(1-\rho)},$$

and $L_q = \lambda W_q = \frac{\lambda^2 E[S^2]}{2(1-\rho)} = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$ since $E[S^2] = \sigma^2 + 1/\mu^2$. $W = W_q + E[S]$ and $L = \lambda W = L_q + \rho$ are the two identities of Proof 3.

(b) $\lambda^2 \sigma^2 = \rho^2 \sigma^2 \mu^2 = \rho^2 c_s^2$; $c_s^2 = 1$ (exponential) returns $M/M/1$, $c_s^2 = 0$ (deterministic) gives $\rho^2/(2(1-\rho))$, half of it.

(c) The chain is not birth–death (a batch of size i jumps $n \rightarrow n+i$), so use the generating function $P(z) = \sum_n p_n z^n$, $Q(z) = \sum_i q_i z^i$. Balance: $\lambda p_0 = \mu p_1$ and, for $n \geq 1$, $(\lambda + \mu)p_n = \mu p_{n+1} + \lambda \sum_{i=1}^n q_i p_{n-i}$. Multiply the n -th equation by z^n and sum ($n \geq 0$, counting the $n=0$ equation as $(\lambda + \mu)p_0 = \mu p_1 + \mu p_0$):

$$(\lambda + \mu)P(z) - \mu p_0 = \frac{\mu}{z}(P(z) - p_0) + \lambda Q(z)P(z) \implies P(z) = \frac{\mu p_0(1-z)}{\mu(1-z) - \lambda z(1-Q(z))}.$$

$P(1) = 1$ fixes p_0 : both numerator and denominator vanish at $z = 1$; by l'Hôpital ($\frac{d}{dz}$ of the denominator at 1 is $-\mu + \lambda Q'(1) = -\mu + \lambda E[X]$), $P(1) = \frac{\mu p_0}{\mu - \lambda E[X]} = 1$, so $p_0 = 1 - \rho$. For $L = P'(1)$ expand around $z = 1 - \varepsilon$: $1 - Q(1 - \varepsilon) = E[X]\varepsilon - \frac{Q''(1)}{2}\varepsilon^2 + O(\varepsilon^3)$ with $Q''(1) = E[X(X - 1)]$, so the denominator is $\varepsilon[(\mu - \lambda E[X]) + \varepsilon \lambda(E[X] + \frac{1}{2}Q''(1))] + O(\varepsilon^3)$ and

$$P(1 - \varepsilon) = \frac{\mu(1 - \rho)}{\mu(1 - \rho) + \varepsilon \lambda(E[X] + \frac{1}{2}E[X(X - 1)])} + O(\varepsilon^2) = 1 - \varepsilon \frac{\lambda(E[X] + E[X^2])}{2\mu(1 - \rho)} + O(\varepsilon^2).$$

Since $P(1 - \varepsilon) = 1 - L\varepsilon + O(\varepsilon^2)$, $L = \frac{\lambda(E[X] + E[X^2])}{2\mu(1 - \rho)} = \frac{\rho + (\lambda/\mu)E[X^2]}{2(1 - \rho)}$. $X \equiv 1$ gives $\frac{2\rho}{2(1 - \rho)}$, the $M/M/1$ value. $\square$

Remarks. (i) Little's law for the bulk queue uses the *customer* rate $\lambda E[X]$: $W = L/(\lambda E[X])$, $L_q = L - \rho$. Airport example: batches of 1, 2, 3, 4 w.p. 0.4, 0.3, 0.2, 0.1, $\lambda = 2/\text{min}$, $\mu = 5/\text{min}$: $E[X] = 2$, $E[X^2] = 5$, $\rho = 0.8$, $L = 7$, $W = 1.75 \text{ min}$, $L_q = 6.2$, $W_q = 1.55 \text{ min}$ — single arrivals at the same customer rate would give $L = 4$. (ii) The web server ($\lambda = \frac{1}{3}/\text{s}$, $E[S] = 2 \text{ s}$, $\sigma = 5 \text{ s}$): $\rho = \frac{2}{3}$, $L_q = 4.83$, $W_q = 14.5 \text{ s}$, $W = 16.5 \text{ s}$, $L = 5.5$; exponential service of the same mean would give $W_q = 4 \text{ s}$. (iii) The textbook route to P-K is the embedded DTMC at departure epochs (the number left behind is Markov); the mean-value argument above is shorter and shows *why* only $E[S^2]$ enters.

(verified in notebook: CP12 §5 (four service laws at $\rho = 2/3$; the heavy-tail warning), §6 (airport batches vs single arrivals))

Checkpoint — answer before moving on.

- Derive $E[R] = \lambda E[S^2]/2$ from the renewal reward theorem of Ch. 7 with cycles = busy cycles, and say exactly where “FCFS” and “PASTA” enter the proof of (a).
- A server can be replaced by a slower but perfectly regular one: $E[S]$ rises by 10%, σ drops from $E[S]$ to 0. At $\rho = 0.8$ (before), does W_q go up or down?
- For the bulk queue, show that $p_0 = 1 - \rho$ also follows from Little's law applied to the server, without any generating function.

AI study card — paste one of these into your AI assistant:

- Ask me to derive the Pollaczek-Khinchine formula by the mean residual service time argument, step by step: what an arrival waits for, the time-average residual service time, and Little's law. Then quiz me on M/D/1 versus M/M/1 at the same rho.
- Set up the balance equations of the $M^*[X]/M/1$ queue and guide me through the generating-function solution: $P(z)$, p_0 from $P(1) = 1$, and L from the expansion at $z = 1$. Make me verify the airport example numbers.

Proof 8 — Jackson networks: traffic equations, the product form, tandem and feedback queues, Braess' paradox, and the optimal split (final-style)

Theorem. Open Jackson network as in §1.6, with $\lambda_i < s_i \mu_i$ at every finite-server station. (a) The traffic equations $\boldsymbol{\lambda} = \boldsymbol{\theta} + \boldsymbol{\lambda}P$ have the unique solution $\boldsymbol{\lambda} = \boldsymbol{\theta}(I - P)^{-1}$. (b) (Product form) $p(\mathbf{x}) = \prod_{i=1}^N \phi_i(x_i)$, where ϕ_i is the stationary law of an $M/M/s_i$ queue with arrival rate λ_i and service rate μ_i . Hence $X_1(t), \dots, X_N(t)$ are independent in steady state, $L = \sum_i L_i$ and $W = L / \sum_i \theta_i$. (c) Tandem ($p_{i,i+1} = 1$): $\lambda_i = \lambda$, $p(\mathbf{x}) = \prod_i (1 - \rho_i) \rho_i^{x_i}$, $W = \sum_i 1/(\mu_i - \lambda)$. (d) Feedback ($N = 1$,

$p_{11} = q$): $L = \frac{\lambda}{\mu(1-q)-\lambda}$, $W_{\text{customer}} = \frac{1}{\mu(1-q)-\lambda}$, $W_{\text{node}} = \frac{1-q}{\mu(1-q)-\lambda}$ per visit. (e) (2024 final Q1) PP(λ) split with probabilities p_1, p_2 to two $M/M/1$ servers with $\mu_1 + \mu_2 = \mu$ fixed: the mean sojourn $\sum_i p_i/(\mu_i - p_i\lambda)$ is minimised by $\mu_i^* = p_i\lambda + \sqrt{p_i} \frac{\mu-\lambda}{\sqrt{p_1}+\sqrt{p_2}}$; for $p_1 = p_2 = \frac{1}{2}$, $\mu_1^* = \mu_2^* = \mu/2$. (f) (Braess) Two routes “queue x then delay 2” and “delay 2 then queue y ”, each queue $\exp(\mu)$, demand PP(2λ): $W = \frac{1}{\mu-\lambda} + 2$. With a free link of delay 1 from x to y and selfish routing, $W = 3$ whenever $\lambda + 1 \leq \mu \leq 2\lambda + 1$ — larger than before if $\mu > \lambda + 1$.

Proof. (a) P is substochastic and from every station a customer eventually leaves (the exit probabilities $r_i = 1 - \sum_j p_{ij}$ reach every state), so $P^n \rightarrow 0$ and $\sum_n P^n = (I - P)^{-1}$ exists: $\lambda = \theta \sum_{n \geq 0} P^n$ — the external rate, plus the rate after one internal transfer, plus two, $\dots$. Every customer visits station i on average $v_i = \lambda_i / \sum_k \theta_k$ times (rate in = customers per unit time $\times$ visits each).

(b) (the Independence handout, completed). Guess $p(\mathbf{x}) = C \prod_i \phi_i(x_i)$ with the $M/M/s_i$ weights $\phi_i(n) = \frac{(\lambda_i/\mu_i)^n}{n!} \phi_i(0)$ ($n < s_i$), $\frac{(\lambda_i/\mu_i)^n}{s_i! s_i^{n-s_i}} \phi_i(0)$ ($n \geq s_i$), so that

$$\frac{\phi_i(n-1)}{\phi_i(n)} = \frac{\min(n, s_i) \mu_i}{\lambda_i}, \quad \frac{\phi_i(n+1)}{\phi_i(n)} = \frac{\lambda_i}{\min(n+1, s_i) \mu_i}.$$

Global balance at $\mathbf{x}$ (rate out = rate in; a completion at i fed back to i changes nothing and is dropped on both sides, so the double sum runs over $j \neq i$):

$$\begin{aligned} \left[\sum_i \theta_i + \sum_i \min(x_i, s_i) \mu_i (1 - p_{ii}) \right] p(\mathbf{x}) &= \sum_i \theta_i p(\mathbf{x} - e_i) + \sum_i \min(x_i + 1, s_i) \mu_i r_i p(\mathbf{x} + e_i) \\ &\quad + \sum_i \sum_{j \neq i} \min(x_i + 1, s_i) \mu_i p_{ij} p(\mathbf{x} + e_i - e_j), \end{aligned}$$

with $p(\cdot) = 0$ outside $\mathbb{Z}_+^N$. Divide by $p(\mathbf{x})$ and use the two ratios: the three sums on the right become

$$\sum_i \theta_i \frac{\min(x_i, s_i) \mu_i}{\lambda_i}, \quad \sum_i \lambda_i r_i = \sum_i \theta_i \quad (\text{traffic equations summed: } \lambda(I - P)\mathbf{e} = \theta\mathbf{e}),$$

$$\begin{aligned} \sum_j \frac{\min(x_j, s_j) \mu_j}{\lambda_j} \sum_{i \neq j} \lambda_i p_{ij} &= \sum_j \frac{\min(x_j, s_j) \mu_j}{\lambda_j} (\lambda_j - \theta_j - \lambda_j p_{jj}) \\ &= \sum_j \min(x_j, s_j) \mu_j (1 - p_{jj}) - \sum_j \theta_j \frac{\min(x_j, s_j) \mu_j}{\lambda_j} \end{aligned}$$

(the j -th traffic equation: $\sum_{i \neq j} \lambda_i p_{ij} = \lambda_j - \theta_j - \lambda_j p_{jj}$). Adding the three, the $\theta_j \min(x_j, s_j) \mu_j / \lambda_j$ terms cancel and what remains is exactly the left side. The guess solves the balance equations, it is summable because each ϕ_i is ($\lambda_i < s_i \mu_i$), and the stationary law of an irreducible CTMC is unique: $p = \prod_i \phi_i$ with $C = 1$, the product of the normalised $M/M/s_i$ laws. Independence, $L = \sum_i L_i$ and, by Little for a box around the network (arrival rate $\sum_i \theta_i$), $W = L / \sum_i \theta_i$ follow.

(c) With $\theta = (\lambda, 0, \dots, 0)$ and $p_{i,i+1} = 1$ the traffic equations give $\lambda_i = \lambda$ (what goes in must come out); (b) gives the product of geometric laws and $L = \sum_i \lambda / (\mu_i - \lambda)$, so $W = L / \lambda = \sum_i 1 / (\mu_i - \lambda)$. Burke’s theorem explains the independence without (b): $X_2(t)$ is a function of the departures from station 1 before t and of station-2 service times, both independent of $X_1(t)$ (Proof 6(b)).

(d) *CTMC view*: the number in the system is birth–death with birth rate λ and death rate $\mu(1 - q)$ (a completion that is fed back does not change the count): an $M/M/1$ with service rate $\mu(1 - q)$, so $L = \frac{\lambda}{\mu(1-q)-\lambda}$, $W_{\text{customer}} = 1/(\mu(1 - q) - \lambda)$. *Network view*: $\hat{\lambda} = \lambda + q\hat{\lambda}$, $\hat{\lambda} = \lambda/(1 - q)$; the node

is $M/M/1(\hat{\lambda}, \mu)$ with $L = \frac{\hat{\lambda}}{\mu - \hat{\lambda}} = \frac{\lambda}{\mu(1-q) - \lambda}$ (the same L) and $W_{\text{node}} = L/\hat{\lambda} = \frac{1-q}{\mu(1-q) - \lambda}$ per visit; a customer makes $1/(1-q)$ visits, so $W_{\text{customer}} = W_{\text{node}}/(1-q)$ — two correct Little's laws with two different arrival rates. The flow into the node (external + fed back) is *not* Poisson; the product form in (b) never needed it to be.

(e) Minimise $f(\mu_1, \mu_2) = \sum_i \frac{p_i}{\mu_i - p_i \lambda}$ subject to $\mu_1 + \mu_2 = \mu$: Lagrange gives $\frac{p_i}{(\mu_i - p_i \lambda)^2} = \kappa$ for both i , i.e. $\mu_i - p_i \lambda = \sqrt{p_i/\kappa}$ — the spare capacity is shared in proportion to $\sqrt{p_i}$, not to p_i . Summing, $\mu - \lambda = \kappa^{-1/2}(\sqrt{p_1} + \sqrt{p_2})$, which gives μ_i^* ; f is convex on the feasible set, so this is the minimum. For $p_1 = p_2 = \frac{1}{2}$ the two servers get $\mu/2$ each.

(f) Before: by Poisson splitting each queue is $M/M/1(\lambda, \mu)$ with sojourn $1/(\mu - \lambda)$, so both routes take $\frac{1}{\mu - \lambda} + 2$. After: write x, y for the mean sojourns at the queues; in a Wardrop equilibrium every used route has the same mean time, $x + 2 = x + 1 + y = 2 + y$, hence $x = y = 1$ and $W = 3$. $x = 1$ means $\frac{1}{\mu - f_x} = 1$, i.e. each queue carries flow $\mu - 1$; the route flows $f_{\text{up}} + f_{\text{mid}} = f_{\text{mid}} + f_{\text{low}} = \mu - 1$ with total 2λ give $f_{\text{mid}} = 2(\mu - 1) - 2\lambda$, $f_{\text{up}} = f_{\text{low}} = 2\lambda - (\mu - 1)$, nonnegative iff $\lambda + 1 \leq \mu \leq 2\lambda + 1$. If $\mu > \lambda + 1$ then $\frac{1}{\mu - \lambda} + 2 < 3$: the link made everyone slower. $\square$

Remarks. (i) The three-station example of the deck: $\theta = (2, 0, 0)$, $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0.3 \\ 1 & 0 & 0 \end{pmatrix}$, $\mu = (4, 3, 2)$: $\lambda_1 = 2 + \lambda_3$, $\lambda_2 = \lambda_1$, $\lambda_3 = 0.3\lambda_2$ give $\lambda = (2.857, 2.857, 0.857)$, $L_i = (2.5, 20, 0.75)$, $L = 23.25$, $W = 11.63$; station 2 at $\rho = 0.95$ holds 20 of the 23 customers. (ii) Bay-Gull Bagels (5 stations, $\theta = (1, 0.2, 0.4, 0, 0)/\text{min}$, $s = (3, 2, \infty, 2, \infty)$): $\lambda = (1, 0.4, 0.741, 1.641, 0.821)$, $L = (6.01, 9.47, 1.48, 5.02, 16.41)$, $L = 38.4$, $W = 24.0$ min. (iii) Where the product form fails: a non-exponential service at a finite-server station (CP13 §2 makes the cashier deterministic: its L drops, the rest barely moves), finite buffers with blocking, state-dependent routing. (iv) In (f) a planner would close the link: with the demand split evenly and nobody on the link the mean sojourn is $\frac{1}{\mu - \lambda} + 2$, the minimum over all routings (CP13 §6). For $\mu > 2 + \sqrt{2}$ (with $\lambda = 1$) the link finally helps: capacity is ample.

(verified in notebook: CP13 §1 (three stations, bottleneck slider), §2 (Bagels vs DES), §3 (tandem), §4 (feedback: W per customer vs per visit), §5 (square-root rule), §6 (Braess: both networks simulated))

Checkpoint — answer before moving on.

- **(final-style)** PP(λ) arrivals join Queue 1 w.p. p_1 and Queue 2 w.p. $1 - p_1$, single servers, $\mu_1 + \mu_2 = \mu$ fixed. Find the μ_1, μ_2 minimising the mean sojourn when $p_1 = \frac{1}{2}$; then for general p_1 . Why is the proportional split $\mu_i = p_i \mu$ not optimal although it equalises the ρ_i ?
- Write the traffic equations of Bay-Gull Bagels from the routing fractions and solve them by hand (three substitutions suffice). Which station fails first if all arrival rates grow by 12%?
- In the feedback queue, which W would a customer complain about and which one appears in the server's log? Verify $W_{\text{customer}} = W_{\text{node}}/(1 - q)$ with $\lambda = 1$, $\mu = 4$, $q = \frac{1}{2}$.
- Finish the product-form verification for $N = 2$, $s_1 = s_2 = 1$, $p_{12} = 1$, $r_2 = 1$ (the tandem queue) by writing out the balance equation at (x_1, x_2) with $x_1, x_2 \geq 1$ and substituting the geometric guess.

AI study card — paste one of these into your AI assistant:

- Give me an open Jackson network with three stations and feedback. Ask me to write the routing matrix, solve the traffic equations, check stability, and compute L_i , W_i per visit, L and W per customer. Then ask why W is not the sum of the W_i .
- Walk me through the verification that the product of $M/M/s_i$ laws solves the global balance equations of a Jackson network: the two ratio identities, the summed traffic equation, and the per-station traffic equation. Let me do each substitution myself.
- Explain Braess' paradox with two parallel queue-plus-delay routes and a new link. Ask me to find the Wardrop equilibrium, the route flows, and the range of μ for which the paradox occurs; then ask what a central planner would do.

A The $G/G/1$ and $G/G/m$ approximations (PLMS handout)

When neither the arrivals nor the service are Markovian there is no closed form. The handout *The $G/G/1$ and $G/G/m$ queueing systems* (PLMS) summarises the usual two-moment approximations. Notation: T inter-arrival time, $E[T] = 1/\lambda$, $C_a^2 = \text{Var}(T)/E[T]^2$; S service time, $E[S] = 1/\mu$, $C_s^2 = \text{Var}(S)/E[S]^2$; $\rho = \lambda/\mu$ ($G/G/1$) or $\lambda/(m\mu)$ ($G/G/m$).

- **$M/G/1$ (exact, Proof 7):** $L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$, $W_q = \frac{\lambda \sigma^2 + \lambda/\mu^2}{2(1-\rho)}$.
- **$G/G/1$, Kingman's heavy-traffic approximation:** $W_q \approx \frac{\rho}{1-\rho} \cdot \frac{C_a^2 + C_s^2}{2} \cdot \frac{1}{\mu}$ — three factors: utilisation, variability, service time. Exact for $M/M/1$ ($C_a^2 = C_s^2 = 1$) and for $M/G/1$ ($C_a^2 = 1$).
- **$G/G/1$, three approximations for L (then $W = L/\lambda$):**

$$\begin{aligned} L &\approx \left(\frac{\rho^2(1 + C_s^2)}{1 + \rho^2 C_s^2} \right) \left(\frac{C_a^2 + \rho^2 C_s^2}{2(1-\rho)} \right) + \rho, \\ L &\approx \left(\frac{\rho(1 + C_s^2)}{2 - \rho + \rho C_s^2} \right) \left(\frac{\rho(2 - \rho)C_a^2 + \rho^2 C_s^2}{2(1-\rho)} \right) + \rho, \\ L &\approx \frac{\rho^2(C_a^2 + C_s^2)}{2(1-\rho)} + \frac{(1 - C_a^2)C_a^2 \rho}{2} + \rho. \end{aligned}$$

- **$G/G/m$:** with $\alpha_m = \frac{\rho^{m+\rho}}{2}$ if $\rho > 0.7$ and $\alpha_m = \rho^{(m+1)/2}$ if $\rho < 0.7$,

$$W_q \approx \frac{\alpha_m}{\mu} \cdot \frac{1}{1-\rho} \cdot \frac{C_a^2 + C_s^2}{2m}, \quad L_q = \lambda W_q, \quad L = L_q + \frac{\lambda}{\mu}.$$

How to use them. (i) They are estimates, not theorems: in CP12 §8 the Poisson-arrival, Gamma-service shop ($\rho = 0.75$, $m = 4$) is within 5% of simulation, while HW9's machine shop (Gamma arrivals with $C_a^2 = 1/16$, constant 1-hour jobs) has $W_q \approx 1.0$ min by the formula and 0.05 min in simulation — L is still 3.05 vs 3.00, because L is dominated by the exact in-service part $\lambda/\mu = 3$. Low variability is where two-moment formulas overestimate most. (ii) All of them share the structure " $M/M/\cdot$ value $\times \frac{C_a^2 + C_s^2}{2}$ ": variability reduction and utilisation are the two levers. (iii) When

the decision matters, simulate (`simulate_queue` with your own samplers) and report a confidence interval. In code: `gg1_approx(lam, mu, ca2, cs2, method)` with `method` $\in \{\text{kingman}, \text{h1}, \text{h2}, \text{h3}\}$ and `ggm_approx(lam, mu, m, ca2, cs2)`.

Where to next. CP13 wrap-up $\rightarrow$ these proofs (gap-fill first) $\rightarrow$ Checkpoint quiz at the start of the next lecture $\rightarrow$ HW 5 Part A (closed-book style, no AI on the first attempt) $\rightarrow$ Part C: how the run length needed to measure L explodes as $\rho \rightarrow 1$, where the product form of the Bagels network breaks, and whether Braess' paradox survives in a discrete-event simulation.